

Antiswing Control for Aerial Transportation of the Suspended Cargo by Dual Quadrotor UAVs

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Abstract—Quadrotor unmanned aerial vehicles, with excellent maneuver performance and flexibility, are widely used in cargo transportation. Due to higher requirements for load capacity, dual quadrotors are expected to work cooperatively so as to transport heavy cargos. However, the aerial transportation system with dual quadrotors exhibits higher degrees of freedom, stronger nonlinearities, and more complex system couplings compared with the single quadrotor transportation system, which brings great challenges to the control of the system. This article focuses on the cooperative control of aerial transportation systems with dual quadrotors. Based on the Lagrange's method, a redundant form of the dynamic model is constructed and presented, where the control inputs are decoupled with system dynamic. According to the cascade characteristics of the system, the inner loop subsystem on the attitude characteristics of the quadrotor, as well as the outer loop subsystem on the cargo's swing motion and the quadrotor's translational motion are defined. Then, a nonlinear hierarchical controller containing relative position constraints and cable tensions compensation is proposed. Specifically, by utilizing the energy-based analysis method, virtual control inputs are designed for the outer loop subsystem. Finally, LaSalle's invariance principle is used to prove the stability of the closed-loop system. Experimental results show satisfactory control performance of the proposed controller.

Index Terms—Aerial transportation system, cooperative control, dual quadrotor unmanned aerial vehicles (UAVs), nonlinear hierarchical control.

I. INTRODUCTION

OWING to the prosperous development of microelectronics technology, electromechanical systems have been widely used in various fields. At present, numerous research results have been carried out on electromechanical systems [1]–[6]. As a type of electromechanical system, quadrotor unmanned aerial vehicles (UAVs) are increasingly being used in many fields with their simple structure, light-weight bodies, and strong mobility. Due to the underactuated feature, strong nonlinearities, and coupling characteristics, the control problems of quadrotor are often complex and challenging. Therefore, more and more research works are carried out on the precise modeling and control for the quadrotor systems [7]–[11].

Aerial transportation systems can deliver cable suspended cargoes, and numerous work has been done on it due to its high research value as well as application prospect. By proving that the aerial transportation system is a differential flat hybrid system, Sreenath *et al.* [12] proposed a nonlinear geometric control scheme to guarantee the tracking of the quadrotor's attitude, load's attitude and position, while Zeng *et al.* [13] presented a path-planning method for reformulating the obstacle avoidance from nonconvex constraints into smooth ones with dual variables. Liang *et al.* [14] presented a nonlinear hierarchical control scheme. A virtual control vector containing saturation functions is designed based on the constructed energy storage function for the outer loop subsystem. Some adaptive control methods [15]–[17] are proposed to tackle the control problems of unknown system parameters, such as the cable length, cargo's mass, and aerodynamic damping coefficients. In addition, the reinforcement learning-based planning method in [18], as well as the neural network-based intelligent control method in [19], is utilized to suppress and eliminate the cargo swing motion.

However, in many cases, the load capacity of a single quadrotor is always limited. Therefore, it is urgent to study the cooperative transportation by dual quadrotor UAVs. In [20] and [21], Pereira *et al.* considered a system composed of a bar tethered to two aerial vehicles, and proposed different control laws for pose tracking and pose stabilization of the bar, respectively. Rossomando *et al.* [22] presented a layered dynamic control structure, which is implemented by a kinematic null-space-based formation controller and a dynamic compensator with a neural sliding-mode controller. In [23], the leader–follower framework is adopted to complete the cooperative transportation of a slender cargo, and three input shapers are introduced to

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reduce the residual oscillations of the follower and the cargo. In addition, many other transportation systems are studied by using inertial measurement units (IMUs) and cameras attached to the quadrotors [24]–[26], which is implemented completely without the feedback information from cargoes and without explicit communication between the collaborating quadrotors. However, the previous work either lacks the validation of actual hardware experiments or does not take into consideration the cargo state feedback.

In this article, a novel hierarchical control scheme is proposed for aerial transportation of the suspended cargo by dual quadrotor UAVs. Our work is carried out on the complete 3-D model with 16 degrees of freedom (DOFs), which is much more challenging compared to the control of the transportation system with just one quadrotor with 8 DOFs [12]–[19], owing to the increased system DOFs and the further enhanced state coupling. Theoretical analyses and experimental results prove that the designed controller drives the quadrotors to their corresponding desired positions accurately with satisfactory cargo swing suppression. Specifically, the dynamic model of the system established by Lagrange's method is divided into two subsystems, i.e., the inner loop subsystem on the attitude characteristics of the quadrotor, and the outer loop subsystem on the cargo's swing motion and the quadrotor's translational motion. For the outer loop, a novel energy storage function is constructed and virtual control vectors with tension compensation is designed for each quadrotor. Then, we derive the thrust inputs as well as the desired attitudes for each quadrotor, and for the inner loop, a geometric tracking controller on the Lie group [8] is utilized to drive the quadrotors to their corresponding desired attitudes. Subsequently, the stability analysis of the closed-loop system is completed by LaSalle's invariance theorem and the lemma about cascade systems. Finally, on the basis of theoretical analysis, we carry out the hardware experiments to verify the feasibility as well as the effectiveness of the proposed controller. Briefly speaking, the main contributions of this article can be summarized as follows:

- 1) When the translation dynamics of one specific quadrotor and the swing angles of the cargo are obtained, the position of the other quadrotor can also be acquired by geometric calculations. Thus, the traditional design idea always establishes the system model based on the independent states, for example, the independent dynamics can be chosen as one quadrotor's translation and rotation, cargo swing, and rotation of the other quadrotor's rotation. While, in this article, system dynamics are analyzed according to redundant states with consideration of both positions of the two quadrotors. This form of redundant expression will facilitate the subsequent controller design and stability analysis as the control inputs are decoupled with system dynamics.
- 2) Based on the characteristics of the cascade system, the controller is designed for the inner and outer loop subsystems, respectively. Specifically, for the outer loop subsystem, a cooperative control scheme is designed by energy-based analysis, and the rope swing angle

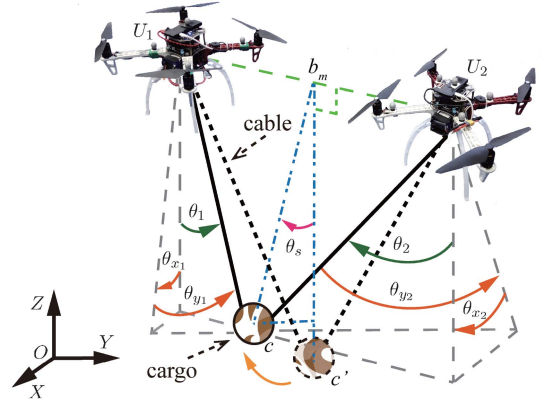


Fig. 1. Aerial transportation of the suspended cargo by dual quadrotor UAVs.

information is introduced into the controller to improve the antiswing performance.

- 3) Barrier Lyapunov functions (BLFs) are introduced to constrain horizontal relative position errors of quadrotors within a predefined range, which avoids accidents caused by quadrotors spacing too large or too small during the flight and improves the security of the system.

The rest of this article is organized as follows. Section II gives the dynamic equations of the aerial transportation system with a suspended cargo by dual quadrotor UAVs and introduces the tracking errors for the outer/inner loop subsystems. In Section III, the energy storage function is constructed and the controller development procedure is illustrated in detail. Subsequently, the corresponding stability analysis is provided in Section IV. Section V presents experimental results. Finally, Section VI concludes this article and future work is summarized.

II. PROBLEM FORMULATION

As Fig. 1 shows, the aerial transportation system with a suspended cargo connected by dual quadrotor UAVs presents complex dynamic characteristics. Point c' indicates the position of the cargo without swinging while point c represents a certain position of the cargo, and point b_m is on the connection line between two quadrotors. Lines $b_m c'$ and $b_m c$ are both perpendicular to $U_1 U_2$. Define the plane formed by lines $U_1 U_2$ and $b_m c'$ as P_q , and define the plane formed by lines $U_1 U_2$ and $b_m c$ as P_l . Therefore, the angle between $b_m c$ and $b_m c'$ is equal to the angle between P_q and P_l , which is defined as the spatial swing angle θ_s depicted by Fig. 1.

Based on the Lagrange's modeling technique, the dynamic equations describing the whole system are provided as follows:

$$M_c(q)\ddot{q} + V_c(q, \dot{q})\dot{q} + G(q) = u \quad (1)$$

$$\dot{R}_i = R_i \hat{\Omega}_i \quad (2)$$

$$J_i \dot{\hat{\Omega}}_i + \hat{\Omega}_i J_i \hat{\Omega}_i = \tau_i, i = 1, 2 \quad (3)$$

where (1) presents the dynamics of the outer loop subsystem, including the quadrotors' translation and the motion of the cargo, and (2) — (3) depict the dynamics of the inner loop subsystem

containing quadrotor rotation motion, with i denoting the i th quadrotor. The operation $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ in (2) is to transform a vector into a skew-symmetric matrix. For the outer loop subsystem, $\mathbf{q} \in \mathbb{R}^{10}$ is the state vector. $M_c(\mathbf{q}), V_c(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{10 \times 10}$, and $\mathbf{G}(\mathbf{q}), \mathbf{u} \in \mathbb{R}^{10}$ denote the inertia matrix, centripetal-Coriolis matrix, gravity vector, and resultant force applied to the outer loop subsystem, respectively. Define g_i as the i th element of the vector $\mathbf{G}(\mathbf{q})$, and the expressions of $\mathbf{q}, \mathbf{u}$, and the entries of $\mathbf{G}(\mathbf{q})$ are as follows:

$$\begin{aligned} \mathbf{q} &= [\xi_1^\top, \xi_2^\top, \Theta_1^\top, \Theta_2^\top]^\top \\ \mathbf{u} &= [\mathbf{u}_1, \mathbf{u}_2, 0, 0, 0, 0]^\top \\ \mathbf{u}_i &= f_i R_i \mathbf{e}_3 - (M_i + 0.5m)g\mathbf{e}_3, \quad \mathbf{e}_3 = [0, 0, 1]^\top \\ g_1 &= g_2 = g_3 = g_4 = g_5 = g_6 = 0 \\ g_7 &= \frac{1}{2}mgl_1 S_{x1} C_{y1}, \quad g_8 = \frac{1}{2}mgl_1 C_{x1} S_{y1} \\ g_9 &= \frac{1}{2}mgl_2 S_{x2} C_{y2}, \quad g_{10} = \frac{1}{2}mgl_2 C_{x2} S_{y2} \end{aligned} \quad (4)$$

where $i = 1, 2$, $\xi_i = [x_i, y_i, z_i]^\top \in \mathbb{R}^3$ denotes the position of each quadrotor's mass center in the inertial reference frame, $\Theta_i = [\theta_{xi}, \theta_{yi}]^\top \in \mathbb{R}^2$ is the projection of the rope swing angles, $M_1, M_2, m, l_1, l_2, g \in \mathbb{R}$, and $f_i \in \mathbb{R}$ denote the quadrotor masses, cargo mass, cable lengths, gravitational acceleration, and applied thrusts, respectively, and $J_i \in \mathbb{R}^{3 \times 3}$, Ω_i , and $\tau_i \in \mathbb{R}^3$ represent the moments of inertia, angular velocities, and moment vector of quadrotors in each body-fixed frame. It can be seen that f_i and τ_i are the actual control inputs to be designed for the entire system. C_{xi}, C_{yi}, S_{xi} , and S_{yi} are abbreviations of $\cos(\theta_{xi}), \cos(\theta_{yi}), \sin(\theta_{xi})$, and $\sin(\theta_{yi})$, respectively, while $C_{x12}, C_{y12}, S_{x12}$, and S_{y12} are abbreviations of $\cos(\theta_{x1} - \theta_{x2}), \cos(\theta_{y1} - \theta_{y2}), \sin(\theta_{x1} - \theta_{x2})$, and $\sin(\theta_{y1} - \theta_{y2})$, respectively. The rope angle θ_i takes the form of $\theta_i = \arccos(C_{xi} C_{yi})$.

As extensively done in literatures on cargo-suspended systems, some reasonable assumptions can be made as follows [14], [27], [28].

Assumption 1: Quadrotors are always above the cargo, i.e.

$$-\pi/2 < \theta_{xi}(t), \theta_{yi}(t) < \pi/2, \quad i = 1, 2 \quad \forall t \geq 0.$$

Property 1: It can be known that $M_c(\mathbf{q})$ is a positive definite symmetric matrix. Moreover, the centripetal-Coriolis matrix $V_c(\mathbf{q}, \dot{\mathbf{q}})$ and the time derivative of $M_c(\mathbf{q})$ satisfy the following skew-symmetric relationship

$$\mathbf{x}^\top \left(\frac{1}{2} \dot{M}_c - V_c \right) \mathbf{x} = 0 \quad \forall \mathbf{x} \in \mathbb{R}^{10}.$$

For convenience in subsequent discussion, we assume that the suspending ropes have same length, the quadrotors have same masses, i.e., $M_1 = M_2 = M$, and the quadrotors are expected to stay at the same height in the end. Our control objective is to drive the quadrotors to move from their initial positions to the desired positions $\xi_{id} = [x_{id}, y_{id}, z_{id}]^\top$, and guarantee that the cargo's spatial swing is suppressed during the flight process,

which can be mathematically described as

$$\xi_1 \rightarrow \xi_{1d}, \quad \xi_2 \rightarrow \xi_{2d}, \quad \theta_s \rightarrow 0. \quad (6)$$

By defining the desired rope swing angles as $\Theta_{id} = [\theta_{xid}, \theta_{yid}]^\top$, the outer loop error signals $\mathbf{e}_q^\top \in \mathbb{R}^{10}$ and $\dot{\mathbf{e}}_q^\top \in \mathbb{R}^{10}$ are defined as

$$\begin{aligned} \mathbf{e}_q &= \mathbf{q} - \mathbf{q}_d = [e_{\xi 1}^\top, e_{\xi 2}^\top, e_{\Theta 1}^\top, e_{\Theta 2}^\top]^\top \\ \dot{\mathbf{e}}_q &= [\dot{e}_{\xi 1}^\top, \dot{e}_{\xi 2}^\top, \dot{e}_{\Theta 1}^\top, \dot{e}_{\Theta 2}^\top]^\top \end{aligned} \quad (7)$$

where $\mathbf{e}_{\xi i}(t) = \xi_i(t) - \xi_{id} = [e_{xi}(t), e_{yi}(t), e_{zi}(t)]^\top \in \mathbb{R}^3$ and $\mathbf{e}_{\Theta i}(t) = \Theta_i(t) - \Theta_{id} = [e_{\theta xi}(t), e_{\theta yi}(t)]^\top \in \mathbb{R}^2$ denote the position errors and the rope swing angle errors, respectively. Then, (1) can be reorganized as

$$\ddot{\mathbf{e}}_q = M_c^{-1} (\mathbf{u} - V_c \dot{\mathbf{e}}_q - \mathbf{G}). \quad (8)$$

For the inner loop subsystem, define the attitude tracking errors $\mathbf{e}_{R1}, \mathbf{e}_{R2} \in \mathbb{R}^3$, and the angular velocity tracking errors $\mathbf{e}_{\Omega 1}, \mathbf{e}_{\Omega 2} \in \mathbb{R}^3$ as

$$\mathbf{e}_{Ri} = \frac{1}{2} \text{vex} (R_{id}^\top R_i - R_i^\top R_{id}) \quad (9)$$

$$\mathbf{e}_{\Omega i} = \Omega_i - R_i^\top R_{id} \Omega_{id} \quad (10)$$

where $\text{vex}(\cdot) : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$ denotes the inverse operation of $\hat{\cdot}$. $R_{id} \in \mathbb{R}^{3 \times 3}$ is the desired attitude and $\Omega_{id} \in \mathbb{R}^3$ represents desired angular velocity of each quadrotor, respectively, whose time derivatives take the form of

$$\dot{\mathbf{e}}_{Ri} = \frac{1}{2} \text{vex} (R_{id}^\top \dot{R}_i \hat{\mathbf{e}}_{\Omega i} + \hat{\mathbf{e}}_{\Omega i} R_i^\top R_{id}) \quad (11)$$

$$J_i \dot{\mathbf{e}}_{\Omega i} = \tau_i - \hat{\Omega}_i J_i \Omega_i + J_i \left(\hat{\Omega}_i R_i^\top R_{id} \Omega_{id} - R_i^\top R_{id} \dot{\Omega}_{id} \right). \quad (12)$$

Hence, the error dynamics can be given by (8), (11), and (12) in a cascade structure.

Remark 1: Actually, the position of a specific quadrotor can be geometrically calculated by the position of the other quadrotor and the rope swing angles, by which the dynamics will present seven DOFs. It can be seen from (1) that the dynamics of the aerial transportation system has ten DOFs, that is, the system dynamics are obtained based on redundant state variables. In the subsequent part, the superiority of the redundant expression in controller design and stability analysis will be presented.

Remark 2: When both quadrotors stay at the same height in the end, $b_m c'$ becomes the midpoint of the two quadrotors and is parallel to the Z -axis of the inertial coordinate system. Define the midpoint of the two quadrotors as $\xi_m = [x_m, y_m, z_m]^\top$ and the cargo position as $\xi_p = [x_p, y_p, z_p]^\top$, which can be calculated geometrically by the quadrotors positions as well as the rope swing angles. Then, the magnitude of the spatial swing θ_s can be obtained by the following formula:

$$\theta_s = \arccos \left(\frac{z_m - z_p}{d_{mp}} \right) \quad (13)$$

where $d_{mp} = \|\xi_m - \xi_p\|_2$ denotes the distance between ξ_m and ξ_p .

Remark 3: Under the condition of (6), the plane composed of the cargo and two quadrotors coincides with the plane P_q , i.e., $x_m = x_p, y_m = y_p$. In this case, the desired rope swing angles θ_{xid} and θ_{yid} can be calculated geometrically as

$$\begin{aligned}\theta_{x1d} &= \arcsin \left[\frac{0.5(x_{2d} - x_{1d})}{\sqrt{l_1^2 - 0.25(y_{2d} - y_{1d})^2}} \right] \\ \theta_{y1d} &= \arcsin \left[\frac{0.5(y_{2d} - y_{1d})}{l_1} \right] \\ \theta_{x2d} &= -\theta_{x1d}, \quad \theta_{y2d} = -\theta_{y1d}.\end{aligned}\quad (14)$$

III. CONTROLLER DEVELOPMENT

This section will detail the design process of the nonlinear hierarchical antiswing controller.

A. Outer Loop Subsystem

The total energy of the outer loop subsystem, containing the kinetic energy and the potential energy, is defined as follows:

$$\begin{aligned}E &= \frac{1}{2} \dot{\mathbf{q}}^\top M_c \dot{\mathbf{q}} + \frac{1}{2} m g l_1 (C_{x1d} C_{y1d} - C_{x1} C_{y1}) \\ &\quad + \frac{1}{2} m g l_2 (C_{x2d} C_{y2d} - C_{x2} C_{y2}).\end{aligned}\quad (16)$$

Based on *Property 1*, the time derivative of $E(t)$ can be calculated as

$$\begin{aligned}\dot{E} &= \dot{\mathbf{q}}^\top \left(M_c \ddot{\mathbf{q}} + \frac{1}{2} \dot{M}_c \dot{\mathbf{q}} \right) + \frac{1}{2} m g l_1 (\dot{\theta}_{x1} S_{x1} C_{y1} + \dot{\theta}_{y1} C_{x1} S_{y1}) \\ &\quad + \frac{1}{2} m g l_2 (\dot{\theta}_{x2} S_{x2} C_{y2} + \dot{\theta}_{y2} C_{x2} S_{y2}) \\ &= \dot{\mathbf{q}}^\top \mathbf{u} - \dot{\mathbf{q}}^\top \mathbf{G}(\mathbf{q}) + \frac{1}{2} \dot{\mathbf{q}}^\top (M_c - 2V_d(\mathbf{q}, \dot{\mathbf{q}})) \dot{\mathbf{q}} \\ &\quad + \frac{1}{2} m g l_1 (\dot{\theta}_{x1} S_{x1} \times C_{y1} + \dot{\theta}_{y1} C_{x1} S_{y1}) \\ &\quad + \frac{1}{2} m g l_2 (\dot{\theta}_{x2} S_{x2} C_{y2} + \dot{\theta}_{y2} C_{x2} S_{y2}) \\ &= \dot{\mathbf{q}}^\top \mathbf{u} = [\dot{e}_{\xi 1}^\top \quad \dot{e}_{\xi 2}^\top]^\top \\ &\quad \times [f_1 R_1 \mathbf{e}_3 - (M + 0.5m) g \mathbf{e}_3 f_2 R_2 \mathbf{e}_3 \\ &\quad - (M + 0.5m) g \mathbf{e}_3]\end{aligned}\quad (17)$$

Subsequently, the following auxiliary energy function is defined as

$$\begin{aligned}E_h &= \frac{1}{2} \mathbf{e}_{\xi 1}^\top K_{p1} \mathbf{e}_{\xi 1} + \frac{1}{2} \mathbf{e}_{\xi 2}^\top K_{p2} \mathbf{e}_{\xi 2} + m g (\mathbf{e}_{\xi 1}^\top \boldsymbol{\rho}_1 + \mathbf{e}_{\xi 2}^\top \boldsymbol{\rho}_2) \\ &\quad + \frac{k_{\sigma x}}{2} \log \frac{\sigma_x^2}{\sigma_x^2 - e_x^2} + \frac{k_{\sigma y}}{2} \log \frac{\sigma_y^2}{\sigma_y^2 - e_y^2}\end{aligned}\quad (18)$$

where $K_{p1} = \text{diag}([k_{px1}, k_{py1}, k_{pz1}])$ and $K_{p2} = \text{diag}([k_{px2}, k_{py2}, k_{pz2}])$ denote positive control gain matrices, and $\boldsymbol{\rho}_1$ and $\boldsymbol{\rho}_2$ are feed-forward coefficients related to the transverse tensions of cables when the quadrotors arrive at the desired positions without cargo swing. Specifically, the mathematical relationship

between cable tensions when the system is in the equilibrium state can be presented as

$$\begin{cases} T_1 S_{1d} = T_2 S_{2d} \\ T_1 C_{1d} + T_2 C_{2d} = m g \end{cases}\quad (19)$$

where T_1 and T_2 are tensions on the cargo along cables l_1 and l_2 , and θ_{1d} and θ_{2d} satisfy

$$\theta_{1d} = \arccos(C_{x1d} C_{y1d}) \quad (20)$$

$$\theta_{2d} = \arccos(C_{x2d} C_{y2d}). \quad (21)$$

Define μ_1 and μ_2 satisfying that $\mu_1 = \frac{T_1}{mg}$ and $\mu_2 = \frac{T_2}{mg}$ and solve (19)–(21), the constant parameters μ_1 and μ_2 can be calculated as

$$\mu_1 = \frac{S_{2d}}{S_{1d} C_{2d} + S_{2d} C_{1d}}, \mu_2 = \frac{S_{1d}}{S_{1d} C_{2d} + S_{2d} C_{1d}}.$$

Subsequently, $\boldsymbol{\rho}_1$ and $\boldsymbol{\rho}_2$ in (18) can be presented as follows:

$$\boldsymbol{\rho}_1 = \mu_1 [C_{y1d} S_{x1d}, S_{y1d}, 0]^\top \quad (22)$$

$$\boldsymbol{\rho}_2 = \mu_2 [C_{y2d} S_{x2d}, S_{y2d}, 0]^\top. \quad (23)$$

In addition, for the vertical tensions of cables, the following equation will be satisfied when the system is stable:

$$\mu_1 C_{y1d} C_{x1d} m g = \mu_2 C_{y1d} C_{x1d} m g = \frac{1}{2} m g. \quad (24)$$

The last two terms in (18) are the BLFs, wherein $k_{\sigma x}$ and $k_{\sigma y}$ are positive constants and $e_x = e_{x1} - e_{x2}$, $e_y = e_{y1} - e_{y2}$ denote the difference of position errors with σ_x and σ_y being the bounds of e_x and e_y , which need to meet the following condition:

$$|e_{x1}(0) - e_{x2}(0)| < \sigma_x, \quad |e_{y1}(0) - e_{y2}(0)| < \sigma_y. \quad (25)$$

By utilizing BLFs, the relative position error of two quadrotors can be limited in a specified range. Define terms $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$ as

$$\boldsymbol{\omega}_1 = \left[\frac{-k_{\sigma x} e_x}{\sigma_x^2 - e_x^2}, \frac{-k_{\sigma y} e_y}{\sigma_y^2 - e_y^2}, 0 \right]^\top, \quad \boldsymbol{\omega}_2 = -\boldsymbol{\omega}_1. \quad (26)$$

To tackle the coupling terms between inner and outer loop subsystems, the thrust term $f_i R_i \mathbf{e}_3$ of each quadrotor can be divided into two parts

$$f_i R_i \mathbf{e}_3 = \boldsymbol{\delta}_i + \mathbf{u}_{id}, i = 1, 2 \quad (27)$$

$$\boldsymbol{\delta}_i = \frac{f_i}{\mathbf{e}_3^\top R_{id}^\top R_i \mathbf{e}_3} [(\mathbf{e}_3^\top R_{id}^\top R_i \mathbf{e}_3) R_i \mathbf{e}_3 - R_{id} \mathbf{e}_3] \quad (28)$$

$$\mathbf{u}_{id} = \frac{f_i}{\mathbf{e}_3^\top R_{id}^\top R_i \mathbf{e}_3} R_{id} \mathbf{e}_3. \quad (29)$$

Next, to separate out the coupling term from the outer loop dynamics, based on (27), the resultant force $\mathbf{u}$ in (1) can be divided into the following forms:

$$\mathbf{u} = \mathbf{f}_v + \mathbf{f}_\delta - \mathbf{f}_g$$

$$\mathbf{f}_v = [\mathbf{u}_{1d}^\top, \mathbf{u}_{2d}^\top, 0, 0, 0, 0]^\top, \mathbf{f}_\delta = [\boldsymbol{\delta}_1^\top, \boldsymbol{\delta}_2^\top, 0, 0, 0, 0]^\top$$

$$\mathbf{f}_g = \left[0, 0, \left(M + \frac{1}{2}m \right) g, 0, 0, \left(M + \frac{1}{2}m \right) g, 0, 0, 0, 0 \right]^\top \quad (30)$$

where $\mathbf{f}_v$ denotes the virtual control input of the outer loop subsystem, $\mathbf{f}_\delta$ represents the coupling between inner and outer loop subsystems, and $\mathbf{f}_g$ is the gravity component vector. Substituting (30) into (8) yields

$$\ddot{\mathbf{e}}_q = M_c^{-1} (\mathbf{f}_v + \mathbf{f}_\delta - \mathbf{f}_g - V_c \dot{\mathbf{e}}_q - \mathbf{G}). \quad (31)$$

In addition, (29) implies that $\mathbf{u}_{id}$ and $R_{id}\mathbf{e}_3$ have the same directions, that is

$$R_{id}\mathbf{e}_3 = \frac{\mathbf{u}_{id}}{\|\mathbf{u}_{id}\|} \quad (32)$$

and the desired unit direction $\mathbf{b}_{3di} \in \mathbb{R}^3$ is defined as $\mathbf{b}_{3di} \equiv R_{id}\mathbf{e}_3$. Substituting (32) into (29) yields

$$\mathbf{u}_{id} = \frac{\mathbf{f}_i \mathbf{u}_{id}}{\mathbf{u}_{id}^\top R_i \mathbf{e}_3} \quad (33)$$

which implies that the real thrust forces f_i of quadrotors can be obtained by the virtual inputs $\mathbf{u}_{id}$ in the following manner:

$$\mathbf{f}_i = \mathbf{u}_{id}^\top R_i \mathbf{e}_3. \quad (34)$$

Combining (16) with (18), a novel energy storage function is constructed as follows:

$$V = E + E_h \quad (35)$$

whose time derivative can be derived as

$$\dot{V} = [\dot{\mathbf{e}}_{\xi 1}^\top \dot{\mathbf{e}}_{\xi 2}^\top] \cdot [\mathbf{h}_1 \mathbf{h}_2]$$

$$\begin{aligned} \mathbf{h}_1 &= f_1 R_1 \mathbf{e}_3 - (M + 0.5m) g \mathbf{e}_3 + K_{p1} \mathbf{e}_{\xi 1} + mg \boldsymbol{\rho}_1 - \boldsymbol{\omega}_1 \\ \mathbf{h}_2 &= f_2 R_2 \mathbf{e}_3 - (M + 0.5m) g \mathbf{e}_3 + K_{p2} \mathbf{e}_{\xi 2} + mg \boldsymbol{\rho}_2 - \boldsymbol{\omega}_2. \end{aligned} \quad (36)$$

Based on (27), $\mathbf{h}_1$ and $\mathbf{h}_2$ can be rewritten as

$$\mathbf{h}_1 = \boldsymbol{\delta}_1 + \mathbf{u}_{1d} - (M + 0.5m) g \mathbf{e}_3 + K_{p1} \mathbf{e}_{\xi 1} + mg \boldsymbol{\rho}_1 - \boldsymbol{\omega}_1 \quad (37)$$

$$\mathbf{h}_2 = \boldsymbol{\delta}_2 + \mathbf{u}_{2d} - (M + 0.5m) g \mathbf{e}_3 + K_{p2} \mathbf{e}_{\xi 2} + mg \boldsymbol{\rho}_2 - \boldsymbol{\omega}_2. \quad (38)$$

The virtual control inputs of the outer loop subsystem can be designed as

$$\begin{aligned} \mathbf{u}_{1d} &= -K_{p1} \mathbf{e}_{\xi 1} - K_{d1} \dot{\mathbf{e}}_{\xi 1} - mg \boldsymbol{\rho}_1 + (M + 0.5m) g \mathbf{e}_3 \\ &\quad + \boldsymbol{\omega}_1 - r_1 K_{s1} \dot{\mathbf{e}}_{\xi 1} \end{aligned} \quad (39)$$

$$\begin{aligned} \mathbf{u}_{2d} &= -K_{p2} \mathbf{e}_{\xi 2} - K_{d2} \dot{\mathbf{e}}_{\xi 2} - mg \boldsymbol{\rho}_2 + (M + 0.5m) g \mathbf{e}_3 \\ &\quad + \boldsymbol{\omega}_2 - r_2 K_{s2} \dot{\mathbf{e}}_{\xi 2} \end{aligned} \quad (40)$$

where $K_{d1} = \text{diag}([k_{dx1}, k_{dy1}, k_{dz1}])$, $K_{d2} = \text{diag}([k_{dx2}, k_{dy2}, k_{dz2}])$, $K_{s1} = \text{diag}([k_{sx1}, k_{sy1}, k_{sz1}])$, and $K_{s2} = \text{diag}([k_{sx2}, k_{sy2}, k_{sz2}])$ are positive control gain matrices, and r_1 and r_2 contain the velocities of the cargo swing angles, that is

$$r_1 = \dot{\theta}_{x1}^2 + \dot{\theta}_{y1}^2 + \dot{\theta}_{x2}^2 + \dot{\theta}_{y2}^2, \quad r_2 = \dot{\theta}_{x1}^2 + \dot{\theta}_{y1}^2 + \dot{\theta}_{x2}^2 + \dot{\theta}_{y2}^2.$$

Substituting (39) and (40) into (36), one can obtain that

$$\begin{aligned} \dot{V} &= \dot{\mathbf{e}}_{\xi 1}^\top [-K_{d1} \dot{\mathbf{e}}_{\xi 1} + \boldsymbol{\delta}_1 - r_1 K_{s1} \dot{\mathbf{e}}_{\xi 1}] + \dot{\mathbf{e}}_{\xi 2}^\top [-K_{d2} \dot{\mathbf{e}}_{\xi 2} + \boldsymbol{\delta}_2 \\ &\quad - r_2 K_{s2} \dot{\mathbf{e}}_{\xi 2}]. \end{aligned} \quad (41)$$

Therefore, based on (32), (39), and (40), the thrust f_i and the desired unit direction $\mathbf{b}_{3di}$ of two quadrotors can be calculated as follows:

$$\begin{aligned} f_i &= \mathbf{u}_{id}^\top R_i \mathbf{e}_3 = [-K_{pi} \mathbf{e}_{\xi i} - K_{di} \dot{\mathbf{e}}_{\xi i} - mg \boldsymbol{\rho}_i - r_i K_{si} \dot{\mathbf{e}}_{\xi i} \\ &\quad + \boldsymbol{\omega}_i + (M + 0.5m) g \mathbf{e}_3]^\top R_i \mathbf{e}_3 \end{aligned} \quad (42)$$

$$\mathbf{b}_{3di} = R_{id} \mathbf{e}_3 = \frac{\mathbf{u}_{id}}{\|\mathbf{u}_{id}\|}. \quad (43)$$

In order to obtain the desired attitude R_{id} , choose an arbitrary vector $\mathbf{b}_{ci} \in \mathbb{R}^3$ not parallel to $\mathbf{b}_{3di}$, and then calculate the following formula:

$$R_{id} = [\mathbf{c}_{2di} \times \mathbf{b}_{3di}; \mathbf{c}_{2di}; \mathbf{b}_{3di}] \quad (44)$$

where $\mathbf{c}_{2di}$ denotes the direction of the body-fixed axis derived by

$$\mathbf{c}_{2di} = \frac{\mathbf{b}_{3di} \times \mathbf{b}_{ci}}{\|\mathbf{b}_{3di} \times \mathbf{b}_{ci}\|}. \quad (45)$$

Based on (7) and (31), the outer loop tracking error vector can be defined as $\mathbf{e} = [\mathbf{e}_q^\top, \dot{\mathbf{e}}_q^\top]^\top \in \mathbb{R}^{20}$, and the outer loop error dynamics is given as

$$\dot{\mathbf{e}} = \boldsymbol{\gamma}(\mathbf{e}, \mathbf{u}_{1d}, \mathbf{u}_{2d}, \boldsymbol{\xi}_{1d}, \boldsymbol{\xi}_{2d}) + \boldsymbol{\epsilon} \quad (46)$$

where $\boldsymbol{\gamma}, \boldsymbol{\epsilon} \in \mathbb{R}^{20}$ can be expressed in detail as

$$\boldsymbol{\gamma} = \Pi_{10} \dot{\mathbf{e}}_q + \Pi_{01} M_c^{-1} (\mathbf{f}_v - \mathbf{f}_g - V \dot{\mathbf{e}}_q - \mathbf{G})$$

$$\boldsymbol{\epsilon} = \Pi_{01} M_c^{-1} \mathbf{f}_\delta$$

where matrices $\Pi_{10}, \Pi_{01} \in \mathbb{R}^{20 \times 10}$ are defined as $\Pi_{10} = [I_{10 \times 10}; 0_{10 \times 10}]$ and $\Pi_{01} = [0_{10 \times 10}; I_{10 \times 10}]$.

Remark 4: In order to achieve control objectives and meet the desired control performance, the designed terms in control inputs (39) and (40) have different effects. Specially, $\boldsymbol{\rho}_1$ and $\boldsymbol{\rho}_2$ are feed-forward coefficients related to the transverse tensions of ropes, $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$ are utilized to limit the relative position errors of two quadrotors, and $-r_1 K_{s1} \dot{\mathbf{e}}_{\xi 1}$ and $-r_2 K_{s2} \dot{\mathbf{e}}_{\xi 2}$ contain the rope swing angular velocities to suppress payload swing during delivery.

B. Inner Loop Subsystem

For inner loop subsystem, the following coordinate-free control approach [8] for each quadrotor is adopted:

$$\begin{aligned} \boldsymbol{\tau}_i &= -K_{Ri} \mathbf{e}_{Ri} - K_{\Omega i} \mathbf{e}_{\Omega i} + \hat{\boldsymbol{\Omega}}_i J_i \boldsymbol{\Omega}_i \\ &\quad - J_i \left(\hat{\boldsymbol{\Omega}}_i R_i^\top R_{id} \boldsymbol{\Omega}_{id} - R_i^\top R_{id} \dot{\boldsymbol{\Omega}}_{id} \right) \end{aligned} \quad (47)$$

where $K_{Ri}, K_{\Omega i} \in \mathbb{R}$ are positive control gains, and the desired angular velocities $\boldsymbol{\Omega}_{id}$ of the quadrotors can be derived by

$$\boldsymbol{\Omega}_{id} = \text{vex} \left(R_{id}^\top \dot{R}_{id} \right). \quad (48)$$

IV. STABILITY AND CONVERGENCE ANALYSIS

This section will present the stability and convergence analysis. First, the coupling term is ignored temporarily and the stability of inner and outer loop subsystems is proved. Then, we will prove that the coupling term satisfies a growth restriction condition, and analyze the stability of the overall system.

A. Stability and Convergence Analysis for the Outer Loop Subsystem

When the coupling term ϵ is temporarily neglected, (46) can be transformed into

$$\dot{e} = \gamma(e, u_{1d}, u_{2d}, \xi_{1d}, \xi_{2d}) \quad (49)$$

for which the following conclusion can be drawn.

Theorem 1: The initial condition (25) as well as the proposed virtual controllers u_{1d} and u_{2d} in (39) and (40) guarantee that the relative position errors e_x and e_y can be bounded by $|e_x| < \sigma_x$ and $|e_y| < \sigma_y$, respectively, and make the equilibrium point of the system asymptotically stable in the sense that

$$\begin{aligned} & \left[e_{\xi 1}^T, e_{\xi 2}^T, \dot{e}_{\xi 1}^T, \dot{e}_{\xi 2}^T, e_{\Theta 1}^T, e_{\Theta 2}^T, \dot{e}_{\Theta 1}^T, \dot{e}_{\Theta 2}^T, \theta_s, \dot{\theta}_s \right]^T \\ &= \left[\mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{2 \times 1}^T, \mathbf{0}_{2 \times 1}^T, \mathbf{0}_{2 \times 1}^T, \mathbf{0}_{2 \times 1}^T, 0, 0 \right]^T. \end{aligned} \quad (50)$$

Proof: According to (41) and the condition that $\delta_1 = \delta_2 = \mathbf{0}_{3 \times 1}$ for system (49), one can conclude that

$$\dot{V} = -\dot{e}_{\xi 1}^T (K_{d1} + r_1 K_{s1}) \dot{e}_{\xi 1} - \dot{e}_{\xi 2}^T (K_{d2} + r_2 K_{s2}) \dot{e}_{\xi 2} \leq 0 \quad (51)$$

which indicates that V is nonincreasing, i.e.

$$V(t) \leq V(0) \ll +\infty \quad (52)$$

where the facts that $|e_x(0)| < \sigma_x$, $|e_y(0)| < \sigma_y$ in (25) are employed.

According to (52), the BLFs in (18) are bounded, that is

$$\frac{k_{\sigma x}}{2} \log \frac{\sigma_x^2}{\sigma_x^2 - e_x^2}, \frac{k_{\sigma y}}{2} \log \frac{\sigma_y^2}{\sigma_y^2 - e_y^2} \in \mathcal{L}_\infty.$$

As the parameters $k_{\sigma x}$, $k_{\sigma y}$, σ_x , and σ_y are constant, one can further conclude that

$$\log \frac{1}{\sigma_x^2 - e_x^2}, \log \frac{1}{\sigma_y^2 - e_y^2} \in \mathcal{L}_\infty \Rightarrow \frac{1}{\sigma_x^2 - e_x^2}, \frac{1}{\sigma_y^2 - e_y^2} \in \mathcal{L}_\infty.$$

If $|e_x(t)|$ tends to exceed the boundary σ_x or $|e_y(t)|$ tends to σ_y , according to (18), there exists sometime making the function $V(t) \rightarrow +\infty$, which conflicts with the conclusion of above-mentioned equations. Hence, it can be concluded that $|e_x| < \sigma_x$, $|e_y| < \sigma_y$, and the relative position constraints ω_1 and ω_2 , are bounded.

Combining (16), (18), and (35), the position errors and swing angle errors are bounded, which can be presented as

$$e_{\xi 1}, e_{\xi 2}, \dot{e}_{\xi 1}, \dot{e}_{\xi 2}, \dot{e}_{\Theta 1}, \dot{e}_{\Theta 2} \in \mathcal{L}_\infty. \quad (53)$$

Based on (53), one can obtain that

$$\dot{\Theta}_1, \dot{\Theta}_2, e_x, e_y, \in \mathcal{L}_\infty \Rightarrow r_1, r_2 \in \mathcal{L}_\infty. \quad (54)$$

According to (39) and (40), it is straightforward to make the following conclusions:

$$u_{1d}, u_{2d} \in \mathcal{L}_\infty. \quad (55)$$

Next, define Φ as

$$\begin{aligned} \Phi = & \left\{ (e_{\xi 1}, e_{\xi 2}, \dot{e}_{\xi 1}, \dot{e}_{\xi 2}, e_{\Theta 1}, e_{\Theta 2}, \dot{e}_{\Theta 1}, \dot{e}_{\Theta 2}, \theta_s, \dot{\theta}_s) \right. \\ & \left. \times | \dot{V}(t) = 0 \right\}. \end{aligned} \quad (56)$$

Let Γ be the largest invariant set in Φ , based on (51), the following conclusions can be achieved:

$$\begin{aligned} \dot{e}_{x1} = \dot{e}_{y1} = \dot{e}_{z1} = \dot{e}_{x2} = \dot{e}_{y2} = \dot{e}_{z2} = 0 \Rightarrow \\ \ddot{e}_{x1} = \ddot{e}_{y1} = \ddot{e}_{z1} = \ddot{e}_{x2} = \ddot{e}_{y2} = \ddot{e}_{z2} = 0, e_{x1} = \beta_{x1} \\ e_{y1} = \beta_{y1}, e_{z1} = \beta_{z1}, e_{x2} = \beta_{x2}, e_{y2} = \beta_{y2}, e_{z2} = \beta_{z2} \end{aligned} \quad (57)$$

with β_{x1} , β_{y1} , β_{z1} , β_{x2} , β_{y2} , and β_{z2} being undetermined constants. The proof will be carried out in two steps.

Step 1: In the invariant set Γ , based on (57), the virtual controller (39) and (40) can be expressed as

$$\begin{aligned} u_{dx1} &= -k_{px1}\beta_{x1} - \mu_1 C_{y1d} S_{x1d} mg - \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \\ u_{dy1} &= -k_{py1}\beta_{y1} - \mu_1 S_{y1d} mg - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \\ u_{dz1} &= -k_{pz1}\beta_{z1} + 0.5mg + Mg \\ u_{dx2} &= -k_{px2}\beta_{x2} - \mu_2 C_{y2d} S_{x2d} mg + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \\ u_{dy2} &= -k_{py2}\beta_{y2} - \mu_2 S_{y2d} mg + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \\ u_{dz2} &= -k_{pz2}\beta_{z2} + 0.5mg + Mg \end{aligned}$$

where u_{dxi} , u_{dyi} , and u_{dzi} denote the components of u . Inserting the virtual control components u_{dxi} , u_{dyi} , and u_{dzi} into (1), and invoking the result in (57), one has that

$$\psi_1 = -k_{px1}\beta_{x1} - \mu_1 C_{y1d} S_{x1d} mg - \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \quad (58)$$

$$\psi_2 = -k_{py1}\beta_{y1} - \mu_1 S_{y1d} mg - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \quad (59)$$

$$\psi_4 = -k_{px2}\beta_{x2} - \mu_2 C_{y2d} S_{x2d} mg + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \quad (60)$$

$$\psi_5 = -k_{py2}\beta_{y2} - \mu_2 S_{y2d} mg + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \quad (61)$$

$$\psi_3 = -k_{pz1}\beta_{z1}, \quad \psi_6 = -k_{pz2}\beta_{z2} \quad (62)$$

$$\eta_1 = \eta_2 = \eta_3 = \eta_4 = 0 \quad (63)$$

where ψ_1 – ψ_6 are defined as

$$\begin{aligned} \psi_1 &= \psi_4 \\ &= \frac{m}{4} l_1 \left(C_{x1} C_{y1} \ddot{\theta}_{x1} - S_{x1} S_{y1} \ddot{\theta}_{y1} - \dot{\theta}_{x1}^2 S_{x1} C_{y1} - 2\dot{\theta}_{x1} \dot{\theta}_{y1} \times \right. \end{aligned}$$

$$C_{x1}S_{y1}-\dot{\theta}_{y1}^2S_{x1}C_{y1})+\frac{m}{4}l_2\left(C_{x2}C_{y2}\ddot{\theta}_{x2}-S_{x2}S_{y2}\ddot{\theta}_{y2}\right. \\ \left.-\ddot{\theta}_{x2}^2S_{x2}C_{y2}-2\dot{\theta}_{x2}\dot{\theta}_{y2}C_{x2}S_{y2}-\dot{\theta}_{y2}^2S_{x2}C_{y2}\right) \quad (64)$$

$$\psi_2 = \psi_5$$

$$= \frac{m}{2}l_1\left(C_{y1}\ddot{\theta}_{y1}-\dot{\theta}_{y1}^2S_{y1}\right)+\frac{m}{2}l_2\left(C_{y2}\ddot{\theta}_{y2}-\dot{\theta}_{y2}^2S_{y2}\right) \quad (65)$$

$$\psi_3 = \psi_6$$

$$= \frac{m}{4}l_1\left(S_{x1}C_{y1}\ddot{\theta}_{x1}+C_{x1}S_{y1}\ddot{\theta}_{y1}-\dot{\theta}_{x1}^2C_{x1}C_{y1}-2\dot{\theta}_{x1}\dot{\theta}_{y1}\right. \\ \left.\times S_{x1}S_{y1}+\dot{\theta}_{y1}^2C_{x1}C_{y1}\right)+\frac{m}{4}l_2\left(S_{x2}C_{y2}\ddot{\theta}_{x2}+C_{x2}S_{y2}\ddot{\theta}_{y2}\right. \\ \left.-\dot{\theta}_{x2}^2C_{x2}C_{y2}-2\dot{\theta}_{x2}\dot{\theta}_{y2}S_{x2}S_{y2}+\dot{\theta}_{y2}^2C_{x2}C_{y2}\right) \quad (66)$$

and η_1 – η_4 are defined as

$$\eta_1 = \frac{m}{4}\left[l_1^2C_{y1}^2\ddot{\theta}_{x1}+l_1l_2C_{x12}C_{y1}C_{y2}\ddot{\theta}_{x2}+l_1l_2S_{x12}C_{y1}S_{y2}\ddot{\theta}_{y2}\right. \\ \left.-2l_1^2\dot{\theta}_{y1}C_{y1}S_{y1}\dot{\theta}_{x1}+l_1l_2\left(\dot{\theta}_{x2}^2S_{x12}C_{y1}C_{y2}-2\dot{\theta}_{x2}\dot{\theta}_{y2}\times\right.\right. \\ \left.\left.C_{x12}C_{y1}S_{y2}+\dot{\theta}_{y2}^2S_{x12}C_{y1}C_{y2}\right)\right]+\frac{1}{2}mgl_1S_{x1}C_{y1} \\ \eta_2 = \frac{m}{4}\left\{l_1^2\ddot{\theta}_{y1}-l_1l_2S_{x12}S_{y1}C_{y2}\ddot{\theta}_{x2}+l_1l_2\ddot{\theta}_{y2}\left(C_{x12}S_{y1}S_{y2}\right.\right. \\ \left.\left.+C_{y1}C_{y2}\right)+l_1^2\dot{\theta}_{x1}^2C_{y1}S_{y1}+l_1l_2\left[2\dot{\theta}_{x2}\dot{\theta}_{y2}S_{x12}S_{y1}S_{y2}\right.\right. \\ \left.\left.+ \dot{\theta}_{x2}^2C_{x12}S_{y1}C_{y2}+\dot{\theta}_{y2}^2\left(C_{x12}S_{y1}C_{y2}-C_{y1}S_{y2}\right)\right]\right\} \quad (67)$$

$$+\frac{1}{2}mgl_1C_{x1}S_{y1} \quad (68)$$

$$\eta_3 = \frac{m}{4}\left[l_1l_2C_{x12}C_{y1}C_{y2}\ddot{\theta}_1-l_1l_2S_{x12}S_{y1}C_{y2}\ddot{\theta}_{y1}+l_2^2C_{y2}^2\ddot{\theta}_{x2}\right. \\ \left.-2l_2^2\dot{\theta}_{y2}C_{y2}S_{y2}\dot{\theta}_{x2}+l_1l_2\left(-\dot{\theta}_{x1}^2S_{x12}C_{y1}C_{y2}-2\dot{\theta}_{x1}\dot{\theta}_{y1}\times\right.\right. \\ \left.\left.C_{x12}S_{y1}C_{y2}-\dot{\theta}_{y1}^2S_{x12}C_{y1}C_{y2}\right)\right]+\frac{1}{2}mgl_2S_{x2}C_{y2} \quad (69)$$

$$\eta_4 = \frac{m}{4}\left\{l_2^2\ddot{\theta}_{y2}+l_1l_2S_{x12}C_{y1}S_{y2}\ddot{\theta}_{x1}+l_1l_2\ddot{\theta}_{y1}\left(C_{x12}S_{y1}S_{y2}\right.\right. \\ \left.\left.+C_{y1}C_{y2}\right)+l_2^2\dot{\theta}_{x2}^2C_{y2}S_{y2}+l_1l_2\left[-\dot{\theta}_{x1}^2S_{x12}S_{y1}S_{y2}\right.\right. \\ \left.\left.+2\dot{\theta}_{x1}\dot{\theta}_{y1}C_{x12}C_{y1}S_{y2}+\dot{\theta}_{y1}^2\left(C_{x12}C_{y1}S_{y2}-S_{y1}C_{y2}\right)\right]\right\} \\ +\frac{1}{2}mgl_2C_{x2}S_{y2}. \quad (70)$$

Furthermore, according to (14) and (15), the relationship between u_{dx1} , u_{dx2} , u_{dy1} , and u_{dy2} can be summarized as

$$u_{dx1}+u_{dx2}=-k_{px1}\beta_{x1}-k_{px2}\beta_{x2} \quad (71)$$

$$u_{dy1}+u_{dy2}=-k_{py1}\beta_{y1}-k_{py2}\beta_{y2}. \quad (72)$$

Based on (71) and (72), adding (58) to (60), and adding (59) to (61) yields

$$\frac{m}{2}l_1\left(C_{x1}C_{y1}\ddot{\theta}_{x1}-S_{x1}S_{y1}\ddot{\theta}_{y1}-\dot{\theta}_{x1}^2S_{x1}C_{y1}-2\dot{\theta}_{x1}\dot{\theta}_{y1}\right. \\ \left.\times C_{x1}S_{y1}-\dot{\theta}_{y1}^2S_{x1}C_{y1}\right)+\frac{m}{2}l_2\left(C_{x2}C_{y2}\ddot{\theta}_{x2}-S_{x2}S_{y2}\ddot{\theta}_{y2}\right. \\ \left.-\dot{\theta}_{x2}^2S_{x2}C_{y2}-2\dot{\theta}_{x2}\dot{\theta}_{y2}C_{x2}S_{y2}-\dot{\theta}_{y2}^2S_{x2}C_{y2}\right)=-k_{px1}\beta_{x1} \\ -k_{px2}\beta_{x2} \quad (73)$$

$$ml_1\left(C_{y1}\ddot{\theta}_{y1}-\dot{\theta}_{y1}^2S_{y1}\right)+ml_2\left(C_{y2}\ddot{\theta}_{y2}-\dot{\theta}_{y2}^2S_{y2}\right)= \\ -k_{py1}\beta_{y1}-k_{py2}\beta_{y2}. \quad (74)$$

The integral of (62), (73), and (74) with respect to time can be presented as

$$l_1\left(C_{x1}C_{y1}\dot{\theta}_{x1}-S_{x1}S_{y1}\dot{\theta}_{y1}\right)+l_2\left(C_{x2}C_{y2}\dot{\theta}_{x2}-S_{x2}S_{y2}\dot{\theta}_{y2}\right) \\ =\frac{2}{m}\left(-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}\right)t+\lambda_x \quad (75)$$

$$l_1C_{y1}\dot{\theta}_{y1}+l_2C_{y2}\dot{\theta}_{y2}=\frac{1}{m}\left(-k_{py1}\beta_{y1}-k_{py2}\beta_{y2}\right)t+\lambda_y, \quad (76)$$

$$l_1\left(S_{x1}C_{y1}\dot{\theta}_{x1}+C_{x1}S_{y1}\dot{\theta}_{y1}\right)+l_2\left(S_{x2}C_{y2}\dot{\theta}_{x2}+C_{x2}S_{y2}\dot{\theta}_{y2}\right) \\ =\frac{4}{m}\left(-k_{pz1}\beta_{z1}\right)t+\lambda_{z1} \quad (77)$$

$$l_1\left(S_{x1}C_{y1}\dot{\theta}_{x1}+C_{x1}S_{y1}\dot{\theta}_{y1}\right)+l_2\left(S_{x2}C_{y2}\dot{\theta}_{x2}+C_{x2}S_{y2}\dot{\theta}_{y2}\right) \\ =\frac{4}{m}\left(-k_{pz2}\beta_{z2}\right)t+\lambda_{z2} \quad (78)$$

where λ_x , λ_y , λ_{z1} , and λ_{z2} are undetermined constants. For equation (75), if $(-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}) \neq 0$, as time approaches infinity, i.e., $t \rightarrow \infty$, the left-hand side of (75) becomes

$$\begin{cases} -\infty, & \text{for } (-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}) < 0 \\ +\infty, & \text{for } (-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}) > 0 \end{cases} \quad (79)$$

which conflicts with the fact that $\dot{\theta}_{x1}, \dot{\theta}_{y1}, \dot{\theta}_{x2}, \dot{\theta}_{y2} \in \mathcal{L}_\infty$. Therefore, the assumption $(-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}) \neq 0$ cannot hold, and one can conclude that

$$-k_{px1}\beta_{x1}-k_{px2}\beta_{x2}=0 \Rightarrow \frac{-k_{px1}}{k_{px2}}\beta_{x1}=\beta_{x2}. \quad (80)$$

Similarly, it is derived that

$$-k_{py1}\beta_{y1}-k_{py2}\beta_{y2}=0 \Rightarrow \frac{-k_{py1}}{k_{py2}}\beta_{y1}=\beta_{y2} \quad (81)$$

$$\beta_{z1}=0, \beta_{z2}=0. \quad (82)$$

Step 2:

Based on the results in (64)–(70), after rigorous mathematical derivation, the following conclusions can be made:

$$l_1C_{x1}C_{y1}\psi_1+l_1S_{x1}C_{y1}\psi_3=\eta_1-\frac{1}{2}mgl_1S_{x1}C_{y1} \quad (83)$$

$$l_1C_{x1}S_{y1}\psi_3-l_1S_{x1}S_{y1}\psi_1+l_1C_{y1}\psi_2=\eta_2-\frac{1}{2}mgl_1C_{x1}S_{y1} \quad (84)$$

$$l_2 C_{x2} C_{y2} \psi_4 + l_4 S_{x2} C_{y2} \psi_6 = \eta_3 - \frac{1}{2} m g l_2 S_{x2} C_{y2} \quad (85)$$

$$l_1 C_{x2} S_{y2} \psi_6 - l_2 S_{x1} S_{y1} \psi_4 + l_1 C_{y1} \psi_5 = \eta_4 - \frac{1}{2} m g l_2 C_{x2} S_{y2}. \quad (86)$$

Substituting (58)–(63) and (82) into (83)–(86), the following formulas related to the virtual control components u_{dxi} , u_{dyi} , and u_{dxi} can be obtained:

$$\begin{aligned} & -\frac{1}{2} m g l_1 S_{x1} C_{y1} \\ & = l_1 C_{x1} C_{y1} \left(-k_{px1} \beta_{x1} - \mu_1 C_{y1} d S_{x1} d m g - \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \right) \end{aligned} \quad (87)$$

$$\begin{aligned} & -\frac{1}{2} m g l_1 C_{x1} S_{y1} \\ & = l_1 S_{x1} S_{y1} \left(k_{px1} \beta_{x1} + \mu_1 C_{y1} d S_{x1} d m g + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \right) \\ & + l_1 C_{y1} \left(-k_{py1} \beta_{y1} - \mu_1 S_{y1} d m g - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right) \end{aligned} \quad (88)$$

$$\begin{aligned} & -\frac{1}{2} m g l_2 S_{x2} C_{y2} \\ & = l_2 C_{x2} C_{y2} \left(-k_{px2} \beta_{x2} - \mu_2 C_{y2} d S_{x2} d m g + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \right) \\ & -\frac{1}{2} m g l_2 C_{x2} S_{y2} \\ & = l_2 S_{x2} S_{y2} \left(k_{px2} \beta_{x2} + \mu_2 C_{y2} d S_{x2} d m g - \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \right) \\ & + l_2 C_{y2} \left(-k_{py2} \beta_{y2} - \mu_2 S_{y2} d m g + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right). \end{aligned} \quad (90)$$

Next, dividing both sides of (87) by $l_1 C_{x1} C_{y1}$ leads to

$$-\frac{1}{2} m g \tan \theta_{x1} = -k_{px1} \beta_{x1} - \mu_1 C_{y1} d S_{x1} d m g - \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2}. \quad (91)$$

By utilizing (24) and (91), after some mathematical manipulation, one can conclude that

$$\frac{1}{2} m g (\tan \theta_{x1} - \tan \theta_{x1d}) = k_{px1} \beta_{x1} + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2}. \quad (92)$$

As for the left-hand side of (92), it is worth noting the relationship between the desired swing angles and the position errors. By combining (14) with (80), the following analysis on (92) can be made:

$$\begin{aligned} \operatorname{sgn} (\tan \theta_{x1} - \tan \theta_{x1d}) &= \operatorname{sgn} (\theta_{x1} - \theta_{x1d}) \\ &= \operatorname{sgn} \left(\arcsin \left(\frac{(x_2 - x_1)}{2l_1} \right) - \arcsin \left(\frac{(x_{2d} - x_{1d})}{2l_1} \right) \right) \\ &= \operatorname{sgn} (x_2 - x_1 - x_{2d} + x_{1d}) \\ &= \operatorname{sgn} (e_{x2} - e_{x1}) \end{aligned}$$

$$= \operatorname{sgn} \left(- \left(1 + \frac{k_{px1}}{k_{px2}} \right) \beta_{x1} \right) = \operatorname{sgn} (-\beta_{x1}). \quad (93)$$

For the right-hand side of (92), according to (80), one can conclude that

$$\begin{aligned} k_{px1} \beta_{x1} + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} &= \beta_{x1} \left(k_{px1} + \frac{k_{\sigma x} \left(1 + \frac{k_{px1}}{k_{px2}} \right)}{\sigma_x^2 - e_x^2} \right) \\ \Rightarrow \operatorname{sgn} \left(k_{px1} \beta_{x1} + \frac{k_{\sigma x} e_x}{\sigma_x^2 - e_x^2} \right) &= \operatorname{sgn} (\beta_{x1}) \end{aligned} \quad (94)$$

which implies that both sides of (92) are opposite and the only solution of (92) is $\beta_{x1} = 0$. Furthermore, by substituting $\beta_{x1} = 0$ into (92), it can be obtained that

$$\theta_{x1} - \theta_{x1d} = 0 \Rightarrow \theta_{x1} = \theta_{x1d}. \quad (95)$$

Substituting (91) into (88), the following equations can be derived:

$$\begin{aligned} & -\frac{1}{2} m g x l_1 C_{x1} S_{y1} = \frac{1}{2} l_1 S_{x1} S_{y1} m g \tan \theta_{x1} \\ & + l_1 C_{y1} \left(-k_{py1} \beta_{y1} - \mu_1 S_{y1} d m g - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right). \end{aligned}$$

For further simplification, $-\frac{1}{2} m g l_1 C_{x1} S_{y1}$ can be extended as $-\frac{m g l_1 C_{x1}^2 S_{y1}}{2 C_{x1}}$, and the following equation can be obtained:

$$\begin{aligned} & -\frac{m g l_1 (C_{x1}^2 + S_{x1}^2) S_{y1}}{2 C_{x1}} = l_1 C_{y1} \left(-k_{py1} \beta_{y1} - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right) \\ & - l_1 \mu_1 C_{y1} S_{y1} d m g \\ \Rightarrow & -\frac{m g l_1 S_{y1}}{2 C_{x1}} = -l_1 \mu_1 C_{y1} S_{y1} d m g \\ & - l_1 C_{y1} \left(k_{py1} \beta_{y1} + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right). \end{aligned} \quad (96)$$

Combining the result of (24), dividing both sides of (96) by $l_1 C_{y1}$ yields

$$\begin{aligned} & -\frac{m g S_{y1}}{2 C_{x1} C_{y1}} = -\mu_1 S_{y1} d m g + \left(-k_{py1} \beta_{y1} - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right) \\ \Rightarrow & -\frac{m g S_{y1}}{2 C_{x1} C_{y1}} = -\frac{m g S_{y1} d}{2 C_{x1} d C_{y1} d} + \left(-k_{py1} \beta_{y1} - \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \right) \\ \Rightarrow & \frac{1}{2} m g \left(\frac{\tan \theta_{y1}}{C_{x1}} - \frac{\tan \theta_{y1d}}{C_{x1d}} \right) = k_{py1} \beta_{y1} + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2}. \end{aligned} \quad (97)$$

By substituting (95) into (97), it can be obtained that

$$\frac{m g}{2 C_{x1d}} (\tan \theta_{y1} - \tan \theta_{y1d}) = k_{py1} \beta_{y1} + \frac{k_{\sigma y} e_y}{\sigma_y^2 - e_y^2} \quad (98)$$

one can derive that $\beta_{y1} = 0$ and $\theta_{y1} = \theta_{y1d}$ via the similar process, as presented in (93)–(95).

Similarly, for (89) and (90), the following conclusions can be achieved:

$$\beta_{x2} = 0, \beta_{y2} = 0, \theta_{x2} = \theta_{x2d}, \theta_{y2} = \theta_{y2d}.$$

According to Remarks 2 and 3, when the rope swing angles are equal to the desired ones, it can be obtained that $x_m = x_p$, $y_m = y_p$, and the spatial angle θ_s is given by

$$\theta_s = \arccos\left(\frac{z_m - z_p}{|z_m - z_p|}\right) = 0. \quad (99)$$

Moreover, the velocity of the spatial angle converges to zero, i.e., $\dot{\theta}_s = 0$.

Hereto, it is known that the largest invariant set Γ only contains the equilibrium point, i.e.

$$\begin{aligned} & \left[e_{\xi 1}^\top, e_{\xi 2}^\top, \dot{e}_{\xi 1}^\top, \dot{e}_{\xi 2}^\top, e_{\theta 1}^\top, e_{\theta 2}^\top, \dot{e}_{\theta 1}^\top, \dot{e}_{\theta 2}^\top, \theta_s, \dot{\theta}_s \right]^\top \\ &= \left[\mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, 0, 0 \right]^\top. \end{aligned}$$

According to LaSalle's invariance theorem [29], the conclusion in (50) can be drawn. Hereto, the proof of *Theorem 1* is completed. ■

B. Stability and Convergence Analysis for the Overall System

For the inner loop subsystem, the detailed analysis of stability is given in [8]. It is proved that the moment control inputs (47) can guarantee that the zero equilibrium of e_{Ri} and $e_{\Omega i}$ is exponentially stable. Because of the existence of the coupling term ϵ in (46), the stability of the overall closed-loop system cannot be guaranteed strictly. Inspired by the theory on cascade systems exploited in [7] and [14], the coupling term will be proved to satisfy a growth restriction condition. First, the following lemma and its proof are presented.

Lemma 1: There exist a positive constant $c_{M\delta}$ and a class- $\mathcal{K}$ function $\Upsilon(\cdot)$ which is differentiable at $[e_{R1}^\top, e_{\Omega 1}^\top, e_{R2}^\top, e_{\Omega 2}^\top]^\top = \mathbf{0}_{12 \times 1}$, such that the coupling term ϵ presents the following property:

$$\|\epsilon\| \leq \Upsilon(e_{R1}, e_{R2}) \|e\|, \quad \text{for } \|e\| \geq c_\delta. \quad (100)$$

Proof: It is known from (54) and (55) that there exists a constant $\varsigma \in \mathbb{R}_+$ satisfying

$$\max(\|\omega_1 - r_1 K_{s1} \dot{e}_{\xi 1}\|, \|\omega_2 - r_2 K_{s2} \dot{e}_{\xi 2}\|) \leq \varsigma.$$

For the quadrotor 1, combining (22) with (39), and exploiting the trigonometric characteristics that $|\cos(\cdot)| \leq 1$, $|\sin(\cdot)| \leq 1$ yields

$$\begin{aligned} \|\mathbf{u}_{1d}\| &= \|-K_{p1} e_{\xi 1} - K_{d1} \dot{e}_{\xi 1} - mg\boldsymbol{\rho}_1 + (0.5m + M)g\mathbf{e}_3 \\ &\quad + \omega_1 - r_1 K_{s1} \dot{e}_{\xi 1}\| \\ &\leq \|K_{p1} e_{\xi 1} + K_{d1} \dot{e}_{\xi 1}\| + \|mg\boldsymbol{\rho}_1\| + (0.5m + M)g + \varsigma \\ &\leq \sqrt{2} \max(\lambda_{p1}, \lambda_{d1}) \|e\| + \left(M + \frac{1 + \sqrt{2}}{2}m\right)g + \varsigma \\ &= \sqrt{2} \max(\lambda_{p1}, \lambda_{d1}) \left[\frac{\left(M + \frac{1 + \sqrt{2}}{2}m\right)g + \varsigma}{\sqrt{2} \max(\lambda_{p1}, \lambda_{d1})} + \|e\| \right] \end{aligned}$$

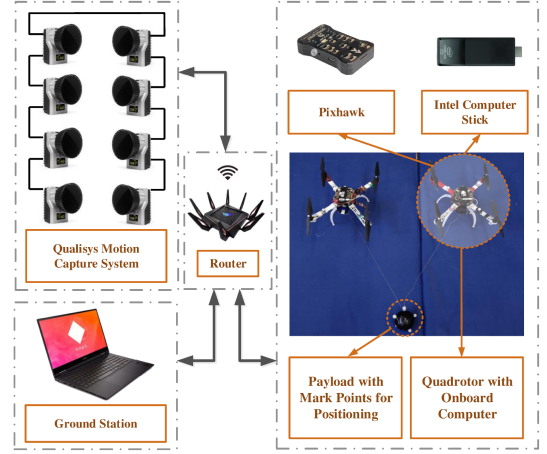


Fig. 2. Hardware experiment platform.

where λ_{p1} and λ_{d1} are the maximum eigenvalues of K_{p1} and K_{d1} , respectively. Define k_{f1} and c_{f1} as

$$k_{f1} = 2\sqrt{2} \max(\lambda_{p1}, \lambda_{d1}), \quad c_{f1} = \frac{\left(M + \frac{1 + \sqrt{2}}{2}m\right)g + \varsigma}{\sqrt{2} \max(\lambda_{p1}, \lambda_{d1})}.$$

Therefore, for quadrotor 1, the following conclusions can be drawn:

$$\|\mathbf{u}_{1d}(e)\| \leq \begin{cases} k_{f1} \|e\|, & \text{for } \|e\| \geq c_{f1} \\ k_{f1} c_{f1}, & \text{for } \|e\| < c_{f1} \end{cases}.$$

It is worth noting that the aforementioned analysis process is only for the quadrotor 1. The same conclusions can be drawn for the quadrotor 2 by defining k_{f2} and c_{f2} as

$$k_{f2} = 2\sqrt{2} \max(\lambda_{p2}, \lambda_{d2}), \quad c_{f2} = \frac{\left(M + \frac{1 + \sqrt{2}}{2}m\right)g + \varsigma}{\sqrt{2} \max(\lambda_{p2}, \lambda_{d2})}$$

where λ_{p2} and λ_{d2} are the maximum eigenvalues of K_{p2} and K_{d2} , respectively. Choosing $c_{M\delta} = \max\{c_{f1}, c_{f2}\}$, one can conclude that

$$\|\mathbf{u}\| \leq \|\mathbf{u}_{1d}\| + \|\mathbf{u}_{2d}\| \leq \begin{cases} (k_{f1} + k_{f2}) \|e\|, & \text{for } \|e\| \geq c_{M\delta} \\ (k_{f1} + k_{f2}) c_{M\delta}, & \text{for } \|e\| < c_{M\delta} \end{cases}.$$

Hereto, for δ_i in (27), after a similar analysis process in [14], we have

$$\begin{aligned} \|\delta_i\| &\leq \|\mathbf{u}_{id}\| \|e_{Ri}\| \leq k_{fi} \|e\| \|e_{Ri}\| \\ &\text{for } \|e\| \geq c_{fi}. \end{aligned} \quad (101)$$

By defining $E_R = \max\{\|e_{R1}\|, \|e_{R2}\|\}$ and combining (46) with (101), the following result is derived:

$$\begin{aligned} \|\epsilon\| &\leq \|\Pi_{01} M_c^{-1}\|_F \|\mathbf{f}_\delta\| \\ &\leq \|\Pi_{01} M_c^{-1}\|_F (\|\delta_1\| + \|\delta_2\|) \\ &\leq \|\Pi_{01} M_c^{-1}\|_F (k_{f1} + k_{f2}) E_R \|e\| \\ &\text{for } \|e\| \geq c_{M\delta} \end{aligned}$$

where $\|\cdot\|_F$ stands for the Frobenius norm. By choosing

$$\Upsilon(e_{R1}, e_{R2}) = \|\Pi_{01} M_c^{-1}\|_F (k_{f1} + k_{f2}) E_R$$

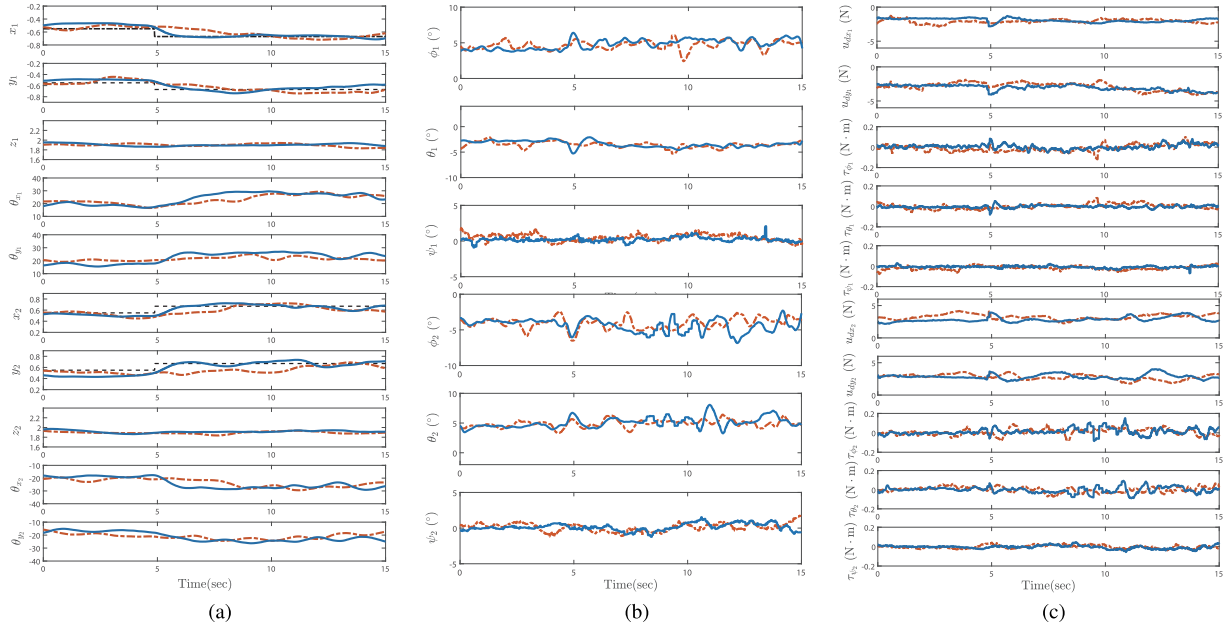


Fig. 3. Results for Experiment 1. (Black dashed lines: Desired positions. Red dotted-dashed lines: PD controller. Blue solid lines: Proposed controller.) (a) Quadrotors positions and cargo swing angles. (b) Quadrotors attitudes. (c) Outer loop control input components in x and y directions and the inner loop control input.

the growth restriction condition in (100) is proven. ■

Theorem 2: The proposed thrust controllers (39) and (40), as well as the moment controller (47) mentioned in Section III, ensure the positioning of both quadrotors with cargo swing suppression and elimination, which can be described as follows:

$$\lim_{t \rightarrow \infty} \begin{bmatrix} e_{\xi 1}^T, e_{\xi 2}^T, \dot{e}_{\xi 1}^T, \dot{e}_{\xi 2}^T, e_{\Theta 1}^T, e_{\Theta 2}^T, \dot{e}_{\Theta 1}^T, \dot{e}_{\Theta 2}^T, \theta_s, \dot{\theta}_s, e_{R1}^T, e_{R2}^T \\ e_{\Omega 1}^T, e_{\Omega 2}^T \end{bmatrix}^T = \begin{bmatrix} \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{2 \times 1}^T, \mathbf{0}_{2 \times 1}^T \\ \mathbf{0}_{2 \times 1}^T, \mathbf{0}_{2 \times 1}^T, 0, 0, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T, \mathbf{0}_{3 \times 1}^T \end{bmatrix}^T.$$

Proof: Combining the analysis process in Section III with the growth restriction condition in Lemma 1, and according to the theorem on the stability of cascade systems [7], the conclusion of Theorem 2 can be obtained. ■

V. EXPERIMENT AND RESULTS

In this section, two groups of experiments are conducted to verify the effectiveness of the proposed method, such as the control performance for cargo swing suppression, cooperative flight performance of the quadrotors, and parametric robustness. In addition, the comparative experiments are carried out by using the PD controller in each group of experiments. The video of experimental records is uploaded in <https://www.youtube.com/watch?v=AOWR3jq6RQs&t=1s>.

As shown in Fig. 2, the hardware experiment platform is composed of dual quadrotor UAVs, Qualisys motion capture system, ground station, and Intel computer sticks. The data including the quadrotor positions, linear velocities, yaw angles, and the payload swing angles are obtained by the Qualisys motion capture and are sent to the ground station based on the local area network. The ground station transmits the control inputs

to the onboard computers via WiFi with 5G band. The onboard computers run the 64-bit Ubuntu 18.04 operating system and are connected to the flight controllers PIXHAWK by mavros-based communication protocol. The entire system runs based on robot operating system.

The physical parameters of the system are given as follows:

$$M_1 = M_2 = 1.43 \text{ kg}, m = 0.152 \text{ kg}$$

$$l_1 = l_2 = 1.32 \text{ m}, g = 9.8 \text{ m/s}^2.$$

The control gains in (39) and (40) are tuned as

$$K_{pi} = \text{diag}([2.0, 4.3, 8.0]), K_{di} = \text{diag}([5.5, 6.3, 12.5])$$

$$K_{si} = \text{diag}([6.0, 6.0, 0.0]).$$

Consider all the initial positions of the quadrotors in the following experiments, the parameters of the cooperative items in (26) are chosen as: $k_{\sigma x} = 6.5, \sigma_x = 2.0, k_{\sigma y} = 6.0$, and $\sigma_y = 2.5$.

The control gains of the PD controller are set as: $k_{pxi} = 3.0, k_{dxi} = 5.8, k_{pyi} = 4.5, k_{dyi} = 6.2, k_{pzi} = 8.0$, and $k_{dzi} = 14.0$.

A. Experiment 1 - Coordinated Flight Verification

This group of experiments mainly tests the cooperative flight ability of the system, which is an important index to evaluate the safety of multiquadrotors system. In practical application scenarios, it is often necessary for quadrotors to quickly perform the relative distance control so as to avoid the collision.

The initial positions and the desired positions of two quadrotors are chosen as

$$\xi_1(0) = [-0.5, -0.5, 2.0]^T \text{ m}, \xi_2(0) = [0.5, 0.5, 2.0]^T \text{ m}$$

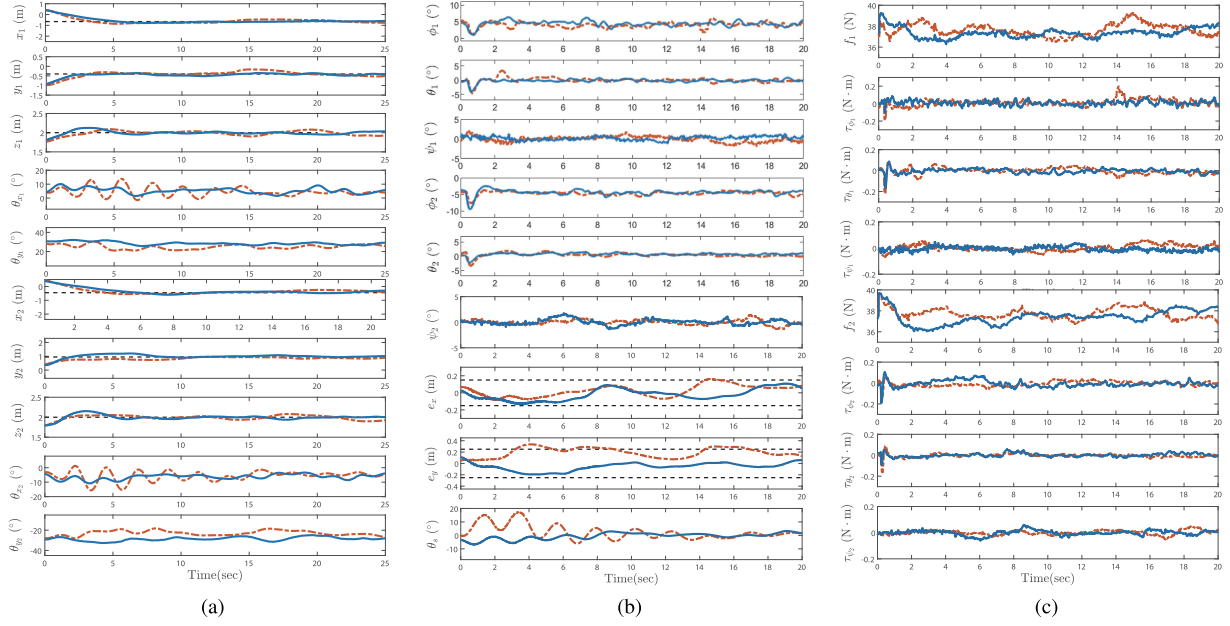


Fig. 4. Results for Experiment 2. (Red dotted-dashed lines: PD controller. Blue solid lines: Proposed controller.) (a) Quadrotors positions and cargo swing angles. (Black-dashed lines: Desired positions.) (b) Quadrotors attitudes, relative errors in the x and y directions, and the spatial swing angle. (Black dashed lines: The Bounds $\pm\sigma_x$ and $\pm\sigma_y$.) (c) Quadrotors control input.

$$\xi_{1d} = [-0.67, -0.67, 2.0]^T \text{ m}, \xi_{2d} = [0.67, 0.67, 2.0]^T \text{ m}.$$

Before $t = 5\text{s}$, both of the quadrotors remain in their initial positions and the PD controller is utilized for the positioning control. The quadrotors switch to the desired positions after $t = 5\text{s}$, while we compare the performance of the proposed controller with that of the PD controller. The results are shown in Fig. 3. As shown in Fig. 3(a), after $t = 5\text{s}$, the proposed method can drive the quadrotors to the desired positions more quickly. Fig. 3(c) indicates that at $t = 5\text{s}$ the control input components u_{dx1} , u_{dy1} , u_{dx2} , and u_{dy2} of the proposed controllers are larger than those of the PD controllers. This is because when the desired distance of the quadrotors increases instantaneously, the denominators of the collaboration terms ω_1 and ω_2 in the proposed controller get smaller which can generate large thrusts to drive the quadrotors away from each other.

B. Experiment 2 - Parametric Robustness Verification

In this group of experiments, the system performs the cargo delivery task. The control performance and the parametric robustness of the proposed controller will be verified. The cargo mass and the cables lengths are chosen as: $l_1 = l_2 = 1.4\text{ m}$, $m = 0.355\text{ kg}$. In addition, for the quadrotors, we change the desired positions as follows:

$$\xi_1(0) = [0.45, -1.0, 1.8]^T \text{ m}, \xi_2(0) = [0.65, 0.4, 1.8]^T \text{ m}$$

$$\xi_{1d} = [-0.65, -0.4, 2.0]^T \text{ m}, \xi_{2d} = [-0.45, 1.0, 2.0]^T \text{ m}.$$

Furthermore, in order to validate the control performance of the proposed method on the relative position between the quadrotors, the bounds of the relative errors e_x and e_y are further reduced to $k_{\sigma x} = 0.5$, $\sigma_x = 0.15$, $k_{\sigma y} = 1.0$, and $\sigma_y = 0.2$.

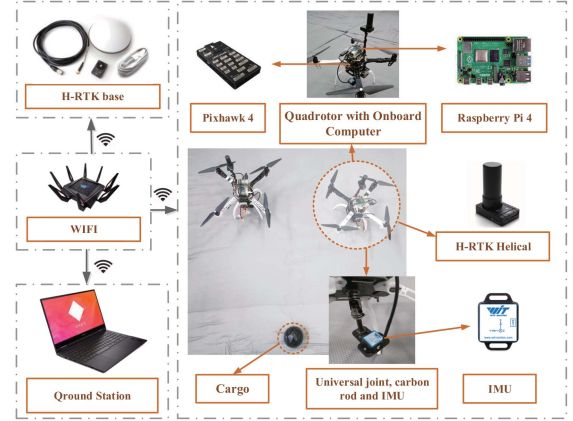


Fig. 5. Hardware configuration of the outdoor environment.

As can be seen from Fig. 4(a), the proposed controller can still guarantee positioning performance and antiswing performance despite changing system parameters, which indicates the excellent parametric robustness of the proposed controller. Fig. 4(b) indicates that the relative errors of the quadrotors driven by the proposed method are constrained within the set range while the comparative PD controller fails to do so. In addition, the spatial swing can also converge to zero more efficiently with the proposed method.

Remark 5: This article focuses on the theoretical derivation of the proposed cooperative control scheme, and we verified the performance of the proposed controller only in an indoor environment with motion capture system. Actually, for the outdoor environment, the proposed control scheme is also applicable,

and the hardware configuration is presented in Fig. 5. The positions of quadrotors are obtained by Holybro H-M8P real-time kinematic differential global positioning system modules, which can achieve centimeter-level positioning accuracy. For each quadrotor, an universal joint is mounted below the quadrotor, and a short carbon rod with IMU is attached to it, which could measure the rope swing angles.

VI. CONCLUSION

This article proposes a novel hierarchical control scheme for aerial transportation of the suspended cargo by dual quadrotor UAVs, which can guarantee the positioning of quadrotors and the swing angle suppression. Specifically, based on the Lagrange's method, a redundant form of the system dynamics model is presented, where the control inputs are decoupled with system dynamics. Then, for the outer loop subsystem, by constructing a new energy storage function, we develop virtual control inputs containing relative position constraints, cable tensions compensation, and the rope swing angles information. LaSalle's invariance theorem is employed to ensure that the equilibrium points are asymptotically stable. Moreover, the coupling terms are proved to satisfy the growth restriction condition and the stability of the overall system is guaranteed by the theory on cascade systems. Experimental results show the superior control performance and parametric robustness of the proposed controller. In ensuing research works, the aerial cooperative transportation system containing more quadrotor UAVs will be studied.

APPENDIX

In this appendix, the detailed expressions of $M_c(\mathbf{q})$, $V_c(\mathbf{q}, \dot{\mathbf{q}})$ will be provided. As the sizes of the matrices $M_c(\mathbf{q})$ and $V_c(\mathbf{q})$ are large, we will use m_{ij} and v_{ij} to represent the $(i+1)$ th row and the $(j+1)$ th column of M_c and V_c , respectively. For M_c , the following expression is given:

$$\begin{aligned} m_{00} &= m_{11} = m_{22} = m_{33} = m_{44} = m_{55} = M + \frac{m}{4} \\ m_{03} &= m_{14} = m_{25} = m_{30} = m_{41} = m_{52} = \frac{m}{4} \\ m_{01} &= m_{02} = m_{04} = m_{05} = m_{10} = m_{12} = m_{13} = m_{15} = 0 \\ m_{16} &= m_{18} = m_{20} = m_{21} = m_{23} = m_{24} = m_{31} = m_{32} = 0 \\ m_{34} &= m_{35} = m_{40} = m_{42} = m_{43} = m_{45} = m_{46} = m_{48} = 0 \\ m_{50} &= m_{51} = m_{53} = m_{54} = m_{61} = m_{64} = m_{67} = m_{76} = 0 \\ m_{81} &= m_{84} = m_{89} = m_{98} = 0, m_{99} = \frac{m}{4} l_2^2 \\ m_{06} &= m_{60} = \frac{l_1 m}{4} C_{x1} C_{y1}, m_{07} = m_{70} = -\frac{l_1 m}{4} S_{x1} S_{y1} \\ m_{08} &= m_{80} = \frac{l_2 m}{4} C_{x2} C_{y2}, m_{09} = m_{90} = -\frac{l_2 m}{4} S_{x2} S_{y2} \\ m_{17} &= m_{71} = \frac{l_1 m}{4} C_{y1}, m_{19} = m_{91} = -\frac{l_2 m}{4} C_{y2} \end{aligned}$$

$$\begin{aligned} m_{26} &= m_{62} = \frac{l_1 m}{4} S_{x1} C_{y1}, m_{27} = m_{72} = \frac{l_1 m}{4} C_{x1} S_{y1} \\ m_{28} &= m_{82} = \frac{l_2 m}{4} S_{x2} C_{y2}, m_{29} = m_{92} = \frac{l_2 m}{4} C_{x2} S_{y2} \\ m_{36} &= m_{63} = \frac{l_1 m}{4} C_{x1} C_{y1}, m_{37} = m_{73} = -\frac{l_1 m}{4} S_{x1} S_{y1} \\ m_{47} &= m_{74} = \frac{l_1 m}{4} C_{y1}, m_{49} = m_{94} = -\frac{l_2 m}{4} C_{y2} \\ m_{56} &= m_{65} = \frac{l_1 m}{4} S_{x1} C_{y1}, m_{57} = m_{75} = \frac{l_1 m}{4} C_{x1} S_{y1} \\ m_{58} &= m_{85} = \frac{l_2 m}{4} S_{x2} C_{y2}, m_{59} = m_{95} = \frac{l_2 m}{4} C_{x2} S_{y2} \\ m_{66} &= \frac{m}{4} l_1^2 C_{y1}^2, m_{68} = m_{86} = \frac{m}{4} l_1 l_2 C_{x12} C_{y1} C_{y2} \\ m_{69} &= m_{96} = \frac{m}{4} l_1 l_2 S_{x12} C_{y1} S_{y2}, m_{77} = \frac{m}{4} l_1^2 \\ m_{78} &= m_{87} = -\frac{m}{4} l_1 l_2 S_{x12} S_{y1} C_{y2}, m_{88} = \frac{m}{4} l_2^2 C_{y2}^2 \\ m_{79} &= m_{97} = \frac{m}{4} l_1 l_2 (C_{x12} S_{y1} S_{y2} + C_{y1} C_{y2}). \end{aligned}$$

For V_c , according to the modeling process, we know that the first six columns are all zeros, i.e., $v_{ij} = 0$, for $i \in [0, 9]$ and $j \in [0, 5]$. The rest elements are given as follows:

$$\begin{aligned} v_{67} &= -v_{76} = -\frac{m}{4} l_1^2 \dot{\theta}_{x1} C_{y1} S_{y1} \\ v_{89} &= -v_{98} = -\frac{m}{4} l_2^2 \dot{\theta}_{x2} C_{y2} S_{y2} \\ v_{16} &= v_{46} = v_{18} = v_{48} = v_{77} = v_{99} = 0 \\ v_{06} &= v_{36} = \frac{l_1 m}{4} (-\dot{\theta}_{x1} S_{x1} C_{y1} - \dot{\theta}_{y1} C_{x1} S_{y1}) \\ v_{07} &= v_{37} = \frac{l_1 m}{4} (-\dot{\theta}_{x1} C_{x1} S_{y1} - \dot{\theta}_{y1} S_{x1} C_{y1}) \\ v_{09} &= v_{39} = \frac{l_2 m}{4} (-\dot{\theta}_{x2} C_{x2} S_{y2} - \dot{\theta}_{y2} S_{x2} C_{y2}) \\ v_{26} &= v_{56} = \frac{l_1 m}{4} (-\dot{\theta}_{x1} C_{x1} C_{y1} - \dot{\theta}_{y1} S_{x1} S_{y1}) \\ v_{27} &= v_{57} = \frac{l_1 m}{4} (-\dot{\theta}_{x1} S_{x1} S_{y1} + \dot{\theta}_{y1} C_{x1} C_{y1}) \\ v_{28} &= v_{58} = \frac{l_2 m}{4} (-\dot{\theta}_{x2} C_{x2} C_{y2} - \dot{\theta}_{y2} S_{x2} S_{y2}) \\ v_{29} &= v_{59} = \frac{l_2 m}{4} (-\dot{\theta}_{x2} S_{x2} S_{y2} + \dot{\theta}_{y2} C_{x2} C_{y2}) \\ v_{66} &= -\frac{m}{4} l_1^2 \dot{\theta}_{y1} C_{y1} S_{y1}, v_{88} = -\frac{m}{4} l_2^2 \dot{\theta}_{y2} C_{y2} S_{y2} \\ v_{17} &= v_{47} = -\frac{l_1 m}{4} \dot{\theta}_{y1} S_{y1}, v_{19} = v_{49} = -\frac{l_2 m}{4} \dot{\theta}_{y2} S_{y2} \\ v_{68} &= \frac{m}{4} l_1 l_2 (\dot{\theta}_{x2} S_{x12} C_{y1} C_{y2} - \dot{\theta}_{y2} C_{x12} C_{y1} S_{y2}) \\ v_{69} &= \frac{m}{4} l_1 l_2 (-\dot{\theta}_{x2} C_{x12} C_{y1} S_{y2} + \dot{\theta}_{y2} S_{x12} C_{y1} C_{y2}) \end{aligned}$$

$$\begin{aligned}
v_{78} &= \frac{m}{4} l_1 l_2 \left(\dot{\theta}_{x2} C_{x12} S_{y1} C_{y2} + \dot{\theta}_{y2} S_{x12} S_{y1} S_{y2} \right) \\
v_{86} &= \frac{m}{4} l_1 l_2 \left(-\dot{\theta}_{x1} S_{x12} C_{y1} C_{y2} - \dot{\theta}_{y1} C_{x12} S_{y1} C_{y2} \right) \\
v_{87} &= \frac{m}{4} l_1 l_2 \left(-\dot{\theta}_{x1} C_{x12} S_{y1} C_{y2} - \dot{\theta}_{y1} S_{x12} C_{y1} C_{y2} \right) \\
v_{96} &= \frac{m}{4} l_1 l_2 \left(\dot{\theta}_{x1} C_{x12} C_{y1} S_{y2} - \dot{\theta}_{y1} S_{x12} S_{y1} S_{y2} \right) \\
v_{79} &= \frac{m}{4} l_1 l_2 \left[\dot{\theta}_{x2} S_{x12} S_{y1} S_{y2} + \dot{\theta}_{22} (C_{x12} S_{y1} C_{y2} - C_{y1} S_{y2}) \right] \\
v_{97} &= \frac{m}{4} l_1 l_2 \left[-\dot{\theta}_{x1} S_{x12} S_{y1} S_{y2} + \dot{\theta}_{y1} (C_{x12} C_{y1} S_{y2} - S_{y1} C_{y2}) \right].
\end{aligned}$$

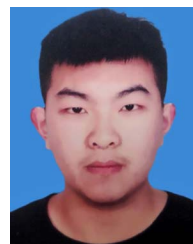
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